

Reconnaissance Trajectory Using OSINT Tools and Google Operators

Internship Assignment

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Reconnaissance Trajectory

Step 1 — Domain Ownership Information

Tool Used: ICANN Lookup (lookup.icann.org)

The first thing I did was try to find out who actually owns the domain. ICANN Lookup gives you **WHOIS data**, which shows basic registration details like:

- Who registered the domain and through which registrar
- Registration and expiry dates
- Name servers linked to the domain

It is a simple first step but it sets the foundation. It tells you who is behind the domain and sometimes even hints at the hosting provider being used.

Step 2 — DNS and Network Intelligence

Tools Used: DNSDumpster, ViewDNS, MXToolbox

Next, I looked into the DNS records of the domain. DNS is basically how the internet knows where to send you when you type in a website address. Using these tools, I could find:

- **A Records** — IP addresses the domain resolves to
- **MX Records** — Mail servers handling email for the domain
- **NS Records** — Name servers managing the domain
- **Subdomains** — e.g., cloud.google.com, drive.google.com

Subdomains are worth paying attention to because they are sometimes overlooked by developers and can accidentally expose internal services or staging environments.

Step 3 — Google Dorking (Search Operators)

Technique: Google Hacking / Google Dorks

This was honestly the part I found most surprising. I did not realize Google could be used this way. **Google Dorks** are advanced search queries that use special operators to filter results in very specific ways. Here are the main ones I studied:

Operator	What It Does
<code>site:</code>	Limits results to a specific website
<code>intitle:</code>	Searches for a keyword in the page title
<code>inurl:</code>	Searches for a keyword in the URL
<code>filetype:</code>	Filters by file type (pdf, xlsx, etc.)
<code>intext:</code>	Searches for text inside the page body
<code>cache:</code>	Shows Google's saved/cached version of a page

Sample Queries:

```
site:example.com filetype:pdf
intitle:"index of"
inurl:login
filetype:xlsx OR filetype:csv
intext:"sql syntax error"
cache:example.com
```

What got me was how these queries can sometimes reveal files or pages that were never meant to be public like an open directory listing, a login panel, or even an error message leaking database details. It made me realize how important it is for organizations to check what Google has indexed from their site.

Step 4 — Internet Exposure Analysis

Tools Used: Shodan, Censys, ZoomEye

I always thought Shodan was some kind of shady hacker tool. Turns out it is more like a search engine, except instead of websites, it indexes **internet-connected devices and services**. Using these tools I could observe:

- Open ports on a server (e.g., port 80, 443, 22)
- Services running on those ports
- Server banners that sometimes expose software version numbers

Outdated software versions showing up here is a big deal — older versions usually have known, documented vulnerabilities that anyone can look up. Security teams use tools like Shodan to get an outsider's view of their own systems.

Step 5 — Technology Stack Identification

Tool Used: Wappalyzer (browser extension)

The last step was figuring out what the website is actually built with. Wappalyzer is a browser extension that quietly reads the responses a website sends back and tells you

things like:

- Web server type (Apache, Nginx, etc.)
- CMS being used (WordPress, Drupal, etc.)
- JavaScript frameworks (React, jQuery, etc.)
- Third-party tools (analytics, CDNs, etc.)

This is still completely passive — Wappalyzer just reads what the site already sends to any browser. But knowing the stack helps understand what kind of vulnerabilities might apply.

Reconnaissance Trajectory Summary

The recon process flows in order, from broad to specific:

Step 1: WHOIS / ICANN Lookup — identify domain ownership

Step 2: DNS footprinting — map subdomains and records

Step 3: Google Dorking — surface exposed files and pages

Step 4: Internet exposure analysis — check open ports and services

Step 5: Technology identification — understand the tech stack

Conclusion

Before starting this assignment, I thought OSINT was just simple searching on Google. After completing it, I understood that OSINT is actually a well-planned and structured process that can reveal a lot of information using only publicly available data.

Overall, it improved my interest in reconnaissance, and I plan to practice more using CTF platforms and legal labs to strengthen my OSINT and cybersecurity skills.

Links

- ICANN Lookup: <https://lookup.icann.org>
- DNSDumpster: <https://dnsdumpster.com>
- MXToolbox: <https://mxtoolbox.com>
- Shodan: <https://www.shodan.io>
- Censys: <https://censys.io>
- Wappalyzer: <https://www.wappalyzer.com>
- ViewDNS: <https://viewdns.info>